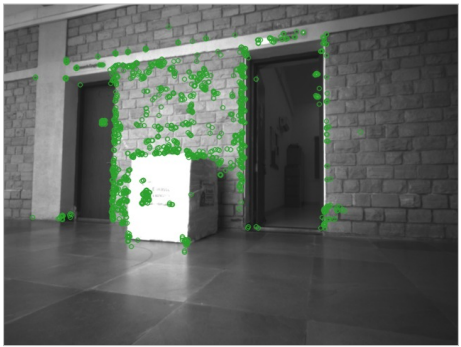
DC PROJECT PRESENTATION

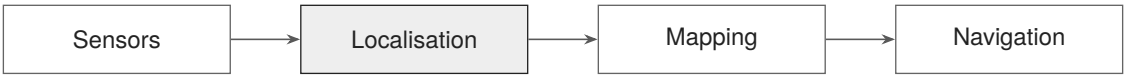
ORB-SLAM3 on Jetson Orin Nano

Real-time visual localisation on an embedded GPU, evaluated against a real robot's ground truth



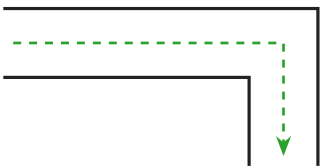
Left IR frame from the Husky recording, with its ORB features

Reliable localisation is the basis of mapping



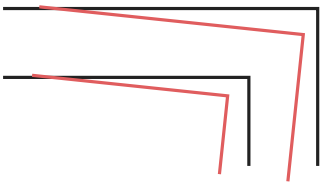
pose error ⇒ map error

(a) Accurate pose



walls line up

(b) Drifting pose

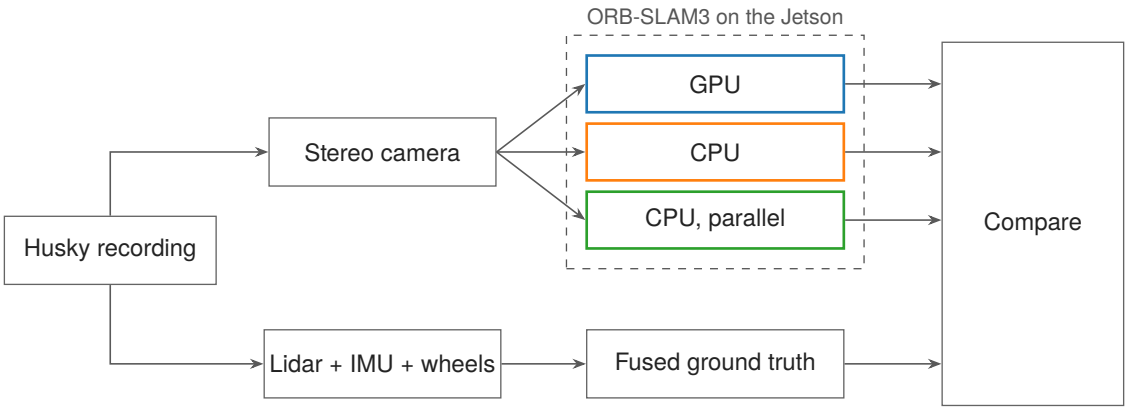


walls doubled, corridor bent

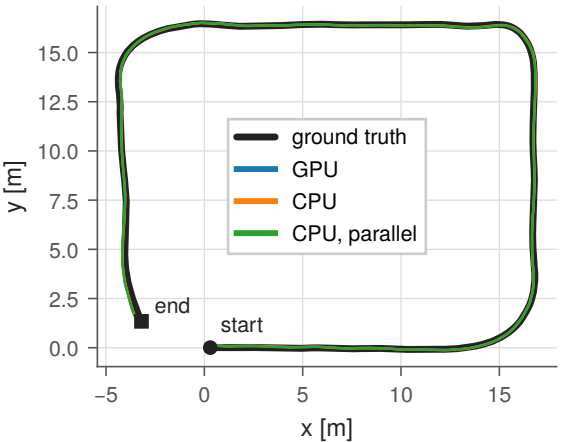
Evaluate Jetson-ORB-SLAM3 on real hardware with real robot data

Question	How we test it	What we measure
GPU as accurate as CPU?	GPU and CPU versions on the same recording, 5 runs each	distance from the true path
Can the pipeline run in real time?	time every processing step of every frame	time per frame vs. time between frames
How much compute does it take?	log the Jetson's power and compute metrics	power, CPU, GPU and memory use
Ablations on sensors and settings?	change one sensor or setting at a time	error and power vs. the default setup

Compare the visual SLAM trajectory with ground truth



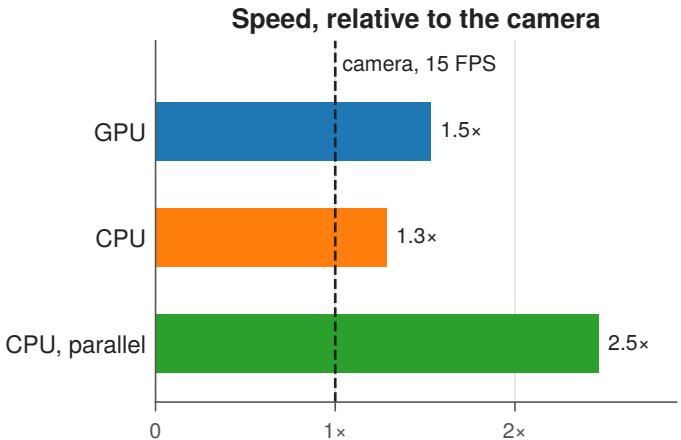
The estimated path follows the ground truth, $\approx 8\,\text{cm}$ average error



The error comes from the scene, not the implementation



Speed, compute and power usage

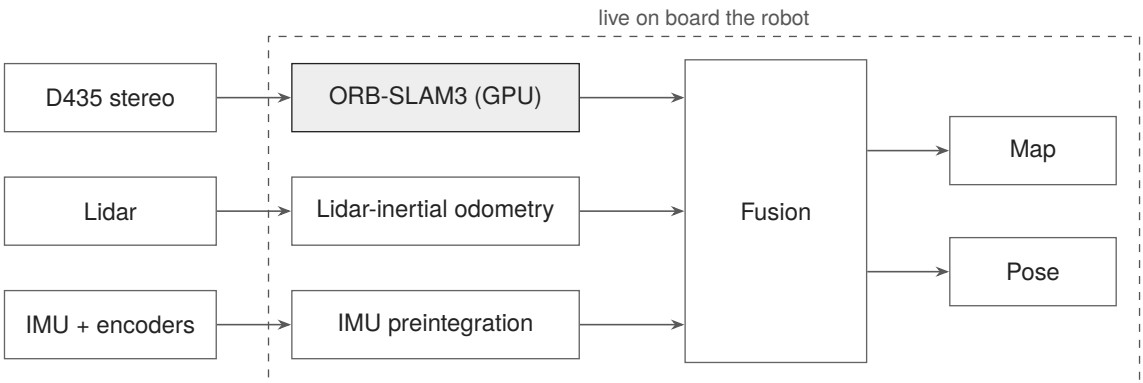


Peak usage	
CPU	61 % of 6 cores
Memory	32 % of 7.6 GB
GPU	31 %
Power	9.9 W

What each change does to pose error, compute and power

Change from the stereo baseline	Pose error	Compute load	Power
Parallel extraction (CPU)	–	↓	–
CPU feature extraction	–	↑	–
Half the frame rate	–	↓	↓
800 instead of 1250 features	–	↓	↓
Add the IMU	↑	–	↓
One camera instead of two	↑↑	↓	↓

Our own platform: ORB-SLAM3 + lidar + IMU



Thank you